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# Paleomagnetic Study of the Juiz de Fora Complex, SE Brazil: Implications for Gondwana

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**Abstract**

The Juiz de Fora Complex is mainly composed of granulites, and granodioritic-migmatite gneisses and is a cratonic basement of the Ribeira belt. Paleomagnetic analysis on samples from 64 sites widely distributed along the Além Paraíba dextral shear zone (SE Brazil, Rio de Janeiro State) yielded a northeastern, steep downward inclination direction (Dm=40.4°, Im=75.4, $\alpha_{95}$=6.0°, K=20.1) for 30 sites. The corresponding paleomagnetic pole (RB) is situated at 335.2°E; 0.6°S ($\alpha_{95}$=10.0°; K=7.9). Rock magnetism indicates that both (titano)magnetite and titanohematite are the main magnetic minerals responsible for this direction. Anisotropy of low-field magnetic susceptibility (AMS) measurements were used to correct the ChRM directions and consequently its corresponding paleomagnetic pole. This correction yielded a new mean ChRM (Dm = 2.9°, Im = 75.4°, $\alpha_{95}$ = 6.4°, K = 17.9) whose paleomagnetic pole RBc is located at 320.1°E, 4.2° N ($\alpha_{95}$=10.3°, K=7.5). Both mean ChRM and paleomagnetic pole obtained from uncorrected and corrected data are statistically different at the 95% confidence circle. Geological and geochronological data suggest that the age of the Juiz de Fora Complex pole is probably between 535–500 Ma, and paleomagnetic results permit further constraint on these ages to the interval 520–500 Ma by comparison with high quality paleomagnetic poles in the 560–500 Ma Gondwana APW path.

**Key words:** Paleomagnetism, correction of ChRM for AMS effect, high-grade metamorphic rocks, Juiz de Fora Complex, Brasiliano orogeny.

## Introduction

It is widely accepted that crust was aggregated into a Supercontinent during Meso-Neoproterozoic times. Although many names have been proposed for this Supercontinent the name Rodinia is accepted by the majority of the researchers nowadays. The life cycle of the Rodinia Supercontinent is constrained by a series of Late Meso- to Early Neoproterozoic collisions aggregated under the term Grenvillian, and by its death recorded in the late Neoproterozoic rift and passive margin succession as the Supercontinent broke up (e.g., Dalziel, 1991; Hoffman, 1991, among others). Although the existence of the Rodinia Supercontinent is widely accepted, its paleogeography, fragmentation, and the formation of the Gondwana Supercontinent is the focus of much debate (e.g., Rogers et al., 1995; Meert and Van der Voo, 1997; Dalziel, 1997; Weil et al., 1998; D'Agrella-Filho et al., 1998; Karlstrom et al., 1999; Wingate and Giddings, 2000; Torsvik et al., 2001; Evans et al., 2000; Meert, 2001). This fact is largely dependent on the scarcity of paleomagnetic data from the several tectonic units that compose these Supercontinents.

The assembly of Western Gondwana was characterized by a long period of accretionary and collisional orogenic processes called the Brasiliano/Pan-African orogeny (~600–520 Ma), involving fragments of the Rodinia Supercontinent (e.g., Hoffman, 1991; Dalziel, 1997; Brito Neves et al., 1999). On the other hand, based on paleomagnetic data, it has been suggested that Western Gondwana was already formed at 550–530 Ma (e.g., Meert et al., 1995; D'Agrella-Filho et al., 2000). In South America, the Brasiliano Orogeny resulted in the collision of the Congo-São Francisco, Kalahari, Amazonian and Rio

de La Plata cratons, and other smaller cratonic fragments. The product of late stages of the Brasiliano Orogeny in southeastern Brazil is known as the Ribeira belt (Trompette, 1994; Machado et al., 1996; Trouw et al., 2000).

The Ribeira belt extends more than 1,500 km in a NE-SW trend along the southeastern coast of Brazil (Fig. 1). It is limited by the Luiz Alves craton to the south and both are covered by the Phanerozoic successions of the Paraná Basin; to the north by the São Francisco craton and there is a lateral transition to the Neoproterozoic Araçuaí belt in which the orogen assumes a predominant N-S trend. To the NW, the NE structural trend of the Ribeira belt overprints the previously developed NNW trend of the Neoproterozoic Brasília belt, resulting in a complex interference zone between the two belts (Trouw et al., 2000).

The Ribeira belt shows major subvertical deep crustal shear zones with dextral movement, which record an important transpressional component in the tectonic evolution of the belt (Trouw et al., 2000). These shear zones separate tectonic domains that have different rock associations, structural and metamorphic features. One of the main shear zones in the Ribeira belt is the Além Paraíba dextral shear zone, a large-scale strike-slip fault about 10 km wide with a NE-SW orientation (Fig. 1) that cuts through the north-central portion of the Rio de Janeiro State.

Four tectono-stratigraphic domains have been defined in the central segment of the Ribeira belt (Fig. 1): (i) The Occidental Terrain corresponds to the reworked margin of the São Francisco Craton. Two crustal scale thrust sheets (Andrelândia and Juiz de Fora) were tangentially transported towards the foreland of the São Francisco Craton; (ii) The Paraíba do Sul Klippe consists of the uppermost thrust slice of the belt; (iii) The Oriental Terrain or Costeiro Terrain, is the locus of a magmatic arc (Tupinambá et al., 1998); and (iv) the Cabo Frio Terrain, a small coastal area, whose structural and geochronological data indicate a late docking event of this terrain (Schmitt et al., 1999). We have concentrated our studies on the Juiz de Fora Complex, which crop out along the Além Paraíba shear zone (Fig. 1) around the Três Rios-Natividade Cities (Rio de Janeiro State, SE Brazil).

The purpose of this paper is to present a new paleomagnetic data set for rocks of the Ribeira belt in order to contribute to a better understanding of the assembly of Gondwana. Since the Juiz de Fora Complex is strongly magnetically anisotropic (Raposo and Egydio-Silva, 2001) both characteristic remanent magnetization (ChRM) and paleomagnetic pole were corrected for the effect of magnetic anisotropy (Raposo et al., 2003). This paper shows that high-grade metamorphic rocks are also suitable for paleomagnetic work.

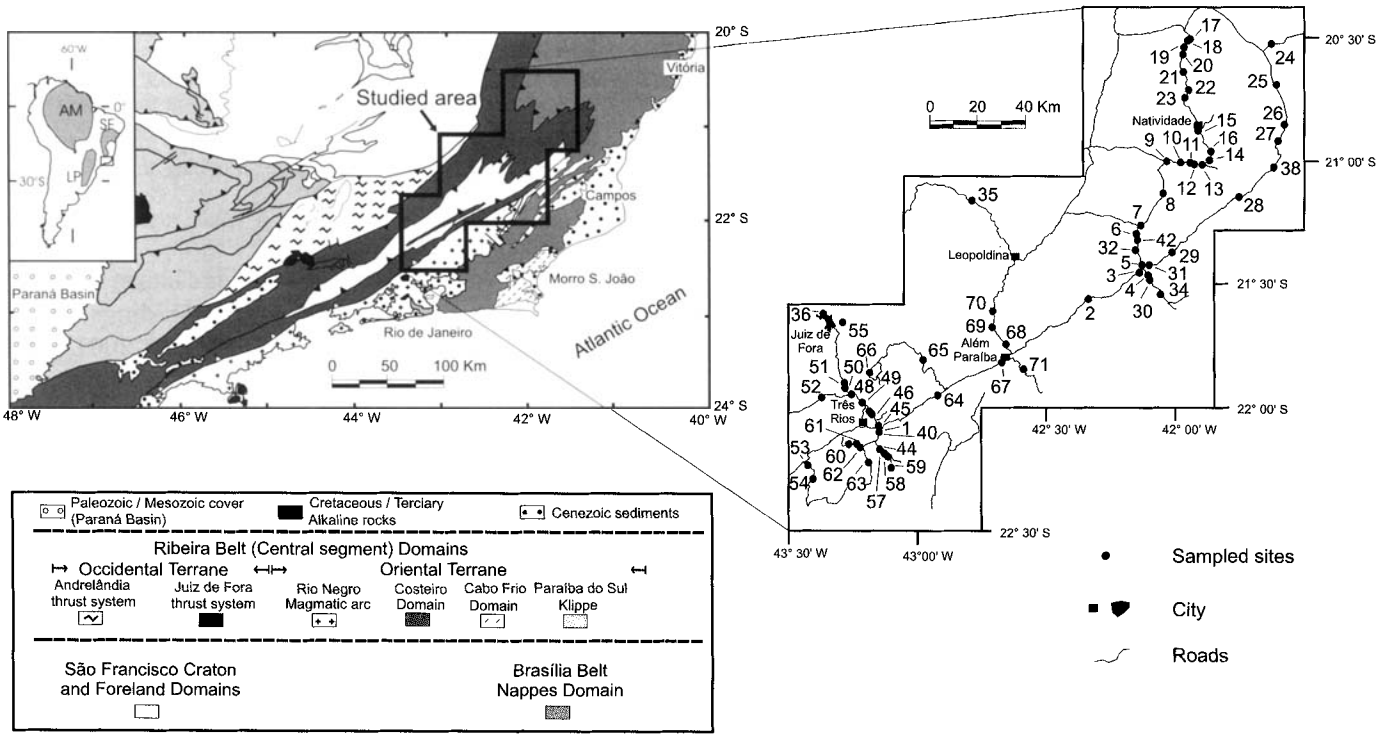



Fig. 1. Main tectono-stratigraphic units of the central segment of the Ribeira belt (modified from Trouw et al., 2000). AM–Amazonia Craton; SF–São Francisco Craton; LP–Rio de la Plata Craton (inset map). Location of sampling sites is shown on the right.

## Geological Setting and Sampling

The central Ribeira belt, in the studied area, is formed by the Andrelândia and the Juiz de Fora Complexes which are crustal-scale thrust sheets, emplaced under amphibolite and granulite facies metamorphic conditions. The Juiz de Fora domain includes three different lithostratigraphic units: (i) pre-1.8 Ga granulitic orthogneisses; (ii) post-1.8 Ga metasedimentary rocks; and (iii) granitoids and charckitoids (Duarte et al., 2000). These rock units are usually imbricated and strongly mylonitized. The Juiz de Fora Complex comprises granulite facies rocks, mainly noritic orthogneisses, garnet-biotite-plagioclase gneisses, amphibolites, pyroxene-plagioclase gneisses, and granodioritic migmatitic gneisses. Recent studies (Duarte et al., 2000; Trouw et al., 2000) have shown that an early metamorphic event is recorded only in the pre-1.8 orthogranulites, probably related to a Paleoproterozoic (Transamazonian) or Mesoproterozoic event. Thermometry data indicate thermal peak conditions around 800–895°C. A later metamorphic event affected the pre- and post-1.8 Ga rocks from the Juiz de Fora domain, and is associated to the main phase of deformation, in which garnet coronas around pyroxenes and metamorphic-related hornblendes record the last stages of this metamorphism. Geothermobarometry, field and petrographic studies indicate that this metamorphic event is marked by temperatures around 700–750°C and pressures of 6–7 kbar. This high-grade metamorphic event is probably related to the syn-collisional stage (565–595 Ma, Trouw et al., 2000; Duarte et al., 2000) of the Brasiliano Orogeny within the Central segment of the Ribeira belt.

The Juiz de Fora Complex is mainly exposed along the Além-Paraíba dextral shear zone (Fig. 1), which was responsible for highly deformed and reworked rocks under the granulite facies conditions at high temperatures (> 800°C, Egydio-Silva at al., 2002) during the late collisional stage of the Brasiliano orogeny. Samples from this region were collected for magnetic analyses. Cylindrical cores (278) from 29 sites were drilled using a portable gasoline-powered rock drill. Normally, 6-10 cores were collected from each outcrop, although, in layered gneissic outcrops, more than 15 cores were taken. Oriented blocks (44) were also collected from 35 sites in a first stage of the field work for a preliminary test. Unfortunately, only one or two samples were collected from each site. Some of these sites were revisited for a more complete sampling (cylindrical cores). Sample orientations were determined using both magnetic and sun compasses, whenever possible.

For magnetic measurements, the sampled cylinders were cut into at least two specimens with a diameter of 2.5 cm and a height of 2.2 cm. At least three specimens of the same size were cut from the block samples.

## Rock Magnetism and Rock-magnetic Properties

Several rock-magnetic experiments were carried out to identify the magnetic carriers responsible for the remanent magnetization and magnetic anisotropies, to obtain information about their paleomagnetic stability and to assess the suitability of the studied samples for paleomagnetic analysis. These experiments included (a) coercivity spectra determined from both anhysteretic remanence and alternating field (AF) demagnetization (Raposo and Egydio-Silva, 2001), (b) measurements of continuous thermomagnetic curves (susceptibility versus (high and low) temperature), hysteresis experiments, and isothermal remanent magnetization (IRM) acquisition. In addition, microscopic investigation were carried out using reflected light, scanning electron microscopy (SEM) (Bascou et al., 2002). Results obtained have shown that there is a mixture of magnetic minerals with predominance of both (titano)magnetite and titanohematite (ilmeno-hematite or hemo-ilmenite), and minor quantity of sulfides such as pyrrhotite (Raposo and Egydio-Silva, 2001).

Since the samples from the Juiz de Fora Complex are strongly heterogeneous at sample scale regarding the amount of magnetic minerals (Raposo and Egydio-Silva, 2001; Raposo et al., 2003), low-field thermomagnetic measurements (K-T curves) under Ar atmospheres were carried out in bulk rock for at least two samples per site using a CS-3 apparatus coupled to the KLY-3 bridge instrument (Agico, Czech Republic). Samples were progressively heated generally up to 700°C and subsequently cooled to room temperature. Alternatively, low-temperature (from about −195°C to room temperature) susceptibility was recorded using a CS3-L apparatus coupled to the KLY-3 bridge instrument. In addition, we obtained low-field thermomagnetic measurements for magnetic mineral concentrates from samples of 23 sites. Representative examples from these experiments are shown in figure 2.

All ferromagnetic mineral concentrations exhibit reversible K-T curves with either no (Fig. 2A) or single (Fig. 2B) Hopkinson peak and a sharp decrease in susceptibility. Generally, a well-defined pick was observed around −150°C that probably indicates the Verwey transition, characteristic of almost pure magnetite. The corresponding high-T susceptibility experiments show a steep decrease in the intensity of susceptibility at temperatures in the ranges of 580°C–590°C (Fig. 2A), 590°C–600°C and around 320°C (Fig. 2B), which could indicate (titano)magnetite and pyrrhotite, respectively.

Magnetic remanence was measured with a JR-5A (sensitivity $\sim 10^{-9}$ Am$^2$; Agico, Czech Republic) and a Molspin (Molspin, Newcastle-upon-Tyne, UK) spinner

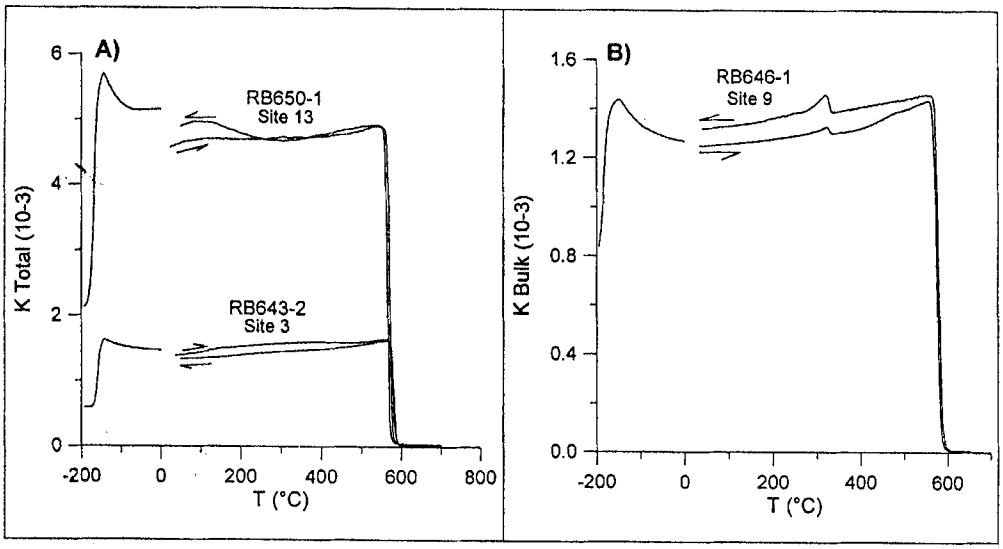



Fig. 2. Representative continuous K-T curves (low and high temperature) on concentrated ferromagnetic mineral separates from the studied rocks. Magnetic susceptibility (K) in SI units.

magnetometers, and a 2G Mountain View (USA) cryogenic magnetometer. The intensity of the natural remanence magnetization (NRM) ranges from very low values of 0.03 mA/m up to 4 A/m. Samples were submitted to stepwise alternating field (AF) demagnetization in a Molspin tumbling demagnetizer (maximum of 100 mT, Molspin, Newcastle-upon-Tyne, UK) by fields from 2.5 or 5 mT up to 100 mT, in steps of 2.5, 5 or 10 mT. Some samples were demagnetized by fields from 2.5 or 5 mT up to 170 mT in the same steps using an automated three-axis AF demagnetizer coupled to the cryogenic magnetometer (maximum of 170 mT). The samples were also submitted to stepwise thermal demagnetization in steps of 50°C up to 500°C and in steps of 20°C up to 700°C, using a non-inductive Schonstedt thermal demagnetizer or a MMTD60 furnace manufactured by Magnetic Measurements. During thermal demagnetization, the low-field susceptibility at room temperature was measured after each step with a SI-2 susceptibility meter (Saphire Instruments).

The coercivity spectra determined from both anhysteretic remanence (Raposo and Egydio-Silva, 2001), and alternating field (AF) demagnetization (Fig. 3A) show that the analyzed samples have both low (< 30 mT) and high coercivity. The thermal demagnetization (Fig. 3B) shows blocking temperatures in the range 550°–690°C, which are coherent with a mixture of (titano)magnetite and titanohematite. These minerals were also observed by optical microscopy. Pyrrhotites, however, were themomagnetically observed in a few samples.

Hysteresis measurements at room temperature were performed on at least three samples per site using a vibrating sample magnetometer (VSM-Nuvo, Molspin, Newcastle-upon-Tyne, UK) in fields up to 1T. Some typical hysteresis curves are reported in Fig. 4. The curves are symmetrical in all cases. Near the origin, contrary to what would be expected, since the samples show different coercivity spectra, no potbellied and wasp-waisted behaviors (Tauxe et al., 1996) were detected. The only exception is site 15 (Fig. 4B) in which only one sample (among 4 analyzed by VSM) shows a slightly wasp-waisted behavior, which probably reflects ferromagnetic phases with different coercivities and/or different ferromagnetic minerals.

The saturation remanent magnetization (Jrs), the saturation magnetization (Js) and coercive force (Hc) were calculated after correction for the paramagnetic contribution. The coercivity of remanence (Hcr) was determined applying progressively increasing backfield after saturation. Unfortunately, all these parameters correspond to a mix of (titano) magnetite, titanohematite and pyrrhotite, and then nothing about magnetic domains can be inferred. Indeed, these parameters were calculated considering that samples reached saturation at 1T (maximum VSM applied field). However, some samples, especially those in which ferromagnetic mineral is dominated by high coercivity fractions, were not able to reach total saturation at the maximum VSM applied field. In fact, isothermal remanent magnetization (IRM) experiments carried out using a pulse magnetometer (MMPM9, Magnetic Measurements) performed on 30 samples, show that some samples as RB649-1F2 (Fig. 5) reached its complete saturation around 1–1.1T while others as RB538-1B2 (Fig. 5) did not reach saturation even in fields as high as 1.6T. The IRM curves also show an increase in IRM acquisition at low fields (< 30 mT) which correspond to (titano)magnetite grains. However, either

no saturation or high field saturation is coherent with the presence of titanohematite, as suggested by coercivity and blocking temperature spectra, and microscopy analysis.

## Paleomagnetic Results

The AF demagnetization was efficient in resolving magnetization components only for a few specimens whereas the thermal demagnetization was better for the majority of them. However, the best results were reached when a combination of both demagnetization methods was applied.

The samples from nearly half of the analyzed sites (34 - most of them are represented by block samples) behaved erratically during both AF and thermal demagnetization and/or no coherent direction could be isolated for them. The specimens from the remaining 30 sites, generally, showed secondary components, probably of viscous origin, which were removed applying 10–20 mT. These secondary components are generally inconsistent even within sites and they are not considered hereafter. The remaining remanent magnetization, in most cases, was removed at temperatures between 450 and 690°C (Fig. 3B) which indicate a mixture of (titano)magnetite and titanohematite as responsible for this magnetization. For these temperatures, demagnetization paths trending towards the origin (Figs. 6–8) permitted the isolation of a single component of magnetization. Characteristic remanent magnetization (ChRM) directions were determined by principal component analysis (Kirschvink, 1980) after selection of linear segments by visual inspection of Zijderveld (1967) plots. Therefore a northeast to southeast, moderate to steep downward inclination magnetization was identified through the least square fit method.

Mean ChRM direction for each site (Table 1; Fig. 9A) was obtained by averaging directions from corresponding specimens, and statistical parameters were calculated assuming a Fisherian distribution (Fisher, 1953). Only sites in which at least three specimens yielded ChRM were considered and presented in table 1. The ChRM directions form a good clustering around the mean Dm=40.4°, Im=75.4° ($\alpha_{95}$=6.0°, K=20.1, N=30). Exception is site 11 whose specimens show a distinct northeast, low downward inclination direction and it was not considered in the mean ChRM direction. The paleomagnetic pole (coded RB) calculated for the Juiz de Fora Complex is located at 335.2°E; 0.6°S ($\alpha_{95}$=10.0°; K=7.9) and was computed as the mean of the virtual geomagnetic poles (VGPs, Table 1) giving unit weight to each site.

## Anisotropy of Low-field Magnetic Susceptibility (AMS) Correction

Since the Juiz de Fora Complex is strongly magnetically anisotropic with an average of ~52% for degree of anisotropy (Raposo and Egydio-Silva, 2001), special attention was given to the influence of the degree of magnetic anisotropy (both AMS and anisotropy of remanent magnetization) on the remanent magnetization vectors.

To determine the amount of deflection in the remanent magnetization directions caused by magnetic anisotropies in the studied rocks, these directions were corrected in all samples (with both AMS and ChRM determinations) from

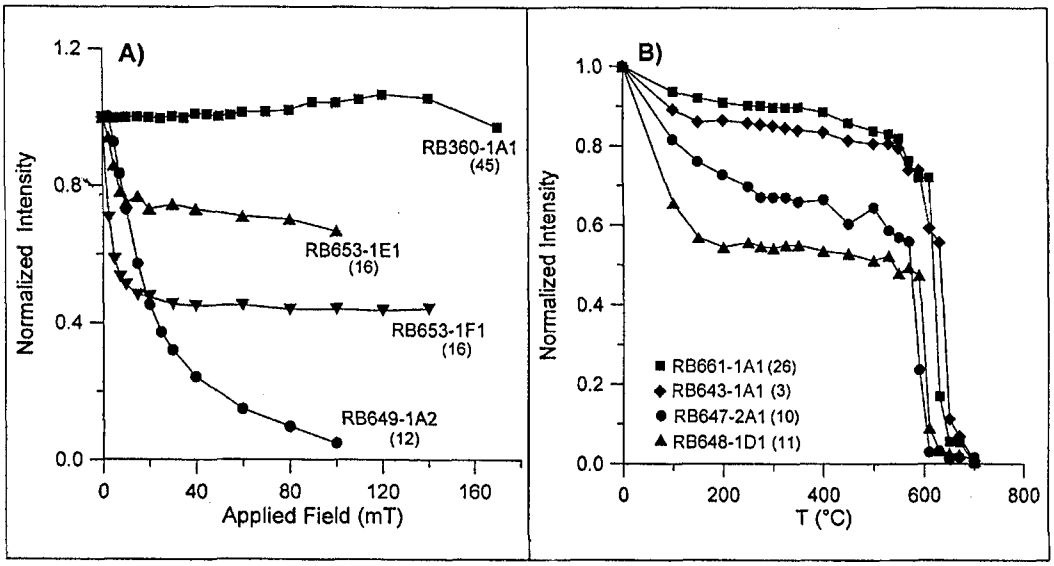



Fig. 3. Normalized intensities versus alternating magnetic field (A) and blocking temperature spectra (B) normalized intensities versus temperature. Data were obtained during the demagnetization processes. Site numbers are indicated in parenthesis.

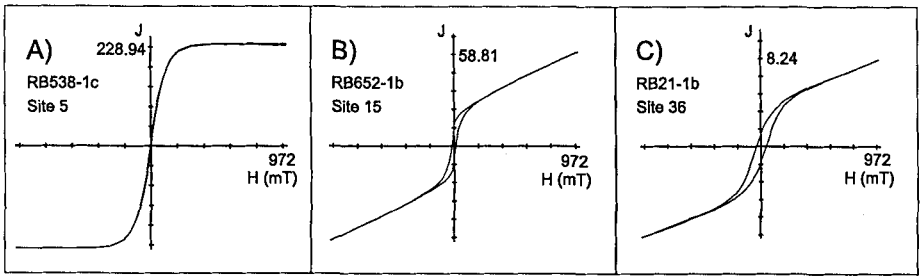



Fig. 4. Typical examples of histeresis curves for the analyzed samples. J = magnetization in $\mu Am^2$, H = applied field.

the sites. The applied method is based on the fundamental equation M=[K]*H, taking into account the direction of the minimum principal axis of susceptibility ($K_{min}$) and the value of the anisotropy degree, and it has widely been discussed by Raposo et al. (2003).

In general, the applied correction produced a little scattering in the directions within each site with respect to uncorrected directions. The corrected directions dislocated towards the minimum principal axis of susceptibility. The correction for the AMS effect (Table 1; Fig. 9B) produced a shift of ~9° between corrected and uncorrected mean ChRM. The paleomagnetic pole calculated from the corrected directions (coded RBc) is around 16° far from the one calculated from the uncorrected directions (RB), and is located at 320.1°E; 4.2°N ($\alpha_{95}$=10.3°, K=7.5; Table 1).

For a few sites uncorrected mean directions have $\alpha_{95}$>20°, however, we did not exclude these sites in the paleomagnetic pole calculation because the exclusion gives a very similar result for both un- and corrected mean ChRM directions.

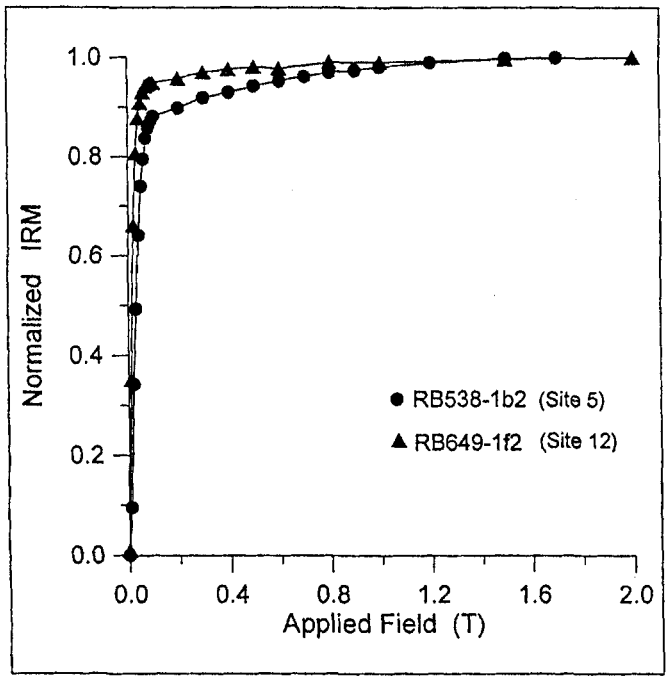



Fig. 5. Examples of IRM acquisition curves for the studied samples. Normalized intensities versus field strength.

## Discussion

The Juiz de Fora Complex was emplaced plastically under mainly granulite facies conditions during the Brasiliano orogeny (~600–520 Ma, Machado et al., 1996). Magnetic anisotropy data (Raposo and Egydio-Silva, 2001) suggest that the ferromagnetic grains and the paramagnetic matrix minerals record the same metamorphic event. Thermometry data suggest that the studied rocks reached temperatures > 800°C (Egydio-Silva et al., 2002) during the late stages of collision (535–520 Ma, Machado el al., 1996). Temperatures as high as these surely indicate that no previous magnetization has survived. Therefore the ChRM direction found to the Juiz de Fora Complex is probably related to the uplift and cooling phase of the last stages of the Brasiliano orogeny in the area. This magnetization is carried by (titano)magnetite and titanohematite. The corresponding paleomagnetic pole, RB, calculated for the Juiz de Fora Complex is located at 335.2°E; 0.6°S ($\alpha_{95}$=10.0°; K=7.9, Table 1). The correction for the AMS effect produced a paleomagnetic pole RBc located at 320.1°E; 4.2°N ($\alpha_{95}$=10.3°, K=7.5, Table 1).

Another correction that could be performed is related to post-tectonic movements (tilting) that may have occurred after the original AMS fabric and ChRM acquisitions. However, there is no evidence that such movements have affected significantly the studied area after the last stage of the Brasiliano orogeny. Indeed, ChRM directions of the Juiz de Fora Complex are similar to those from high-grade metamorphic rocks of the Piquete Complex (also from the Ribeira belt; PQ pole, Fig. 10) situated ~200 km SW of the studied area (D'Agrella-Filho et al., 1986). In addition, paleomagnetic poles for carbonatic rocks from the Una and the Bambuí Groups (São Francisco Craton) are similar (D'Agrella-Filho et al., 2000; poles BGB and BGC, Fig. 10). These carbonatic rocks are more than 1000 km far from each other, and also have only northeast to southeast, steep downward inclined

Table 1. Paleomagnetic results.

| Site | Rock type | Site mean directions | | | | VGP | | AMS corrected site mean directions | | | AMS corrected VGP | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | N/n | Dec. (°) | Inc. (°) | $\alpha_{95}$ (°) | Plong. (°E) | Plat. (°N) | Dec. (°) | Inc. (°) | $\alpha_{95}$ (°) | Plong (°E) | Plat. (°N) |
| 2 | Orthogneiss | 17/12 | 88.5 | 79.9 | 6.1 | 338.6 | -19.8 | 15.5 | 79.8 | 6.0 | 322.9 | -2.5 |
| 3 | granulite | 13/13 | 5.8 | 73.8 | 3.4 | 320.8 | 8.5 | 328.4 | 59.2 | 5.1 | 292.2 | 21.8 |
| 4 | garnet-pyroxene granulite | 6/5 | 64.3 | 65.3 | 8.0 | 355.5 | 0.2 | 26.7 | 70.7 | 8.3 | 333.1 | 10.2 |
| 5 | orthogneiss | 10/10 | 62.3 | 71.8 | 6.4 | 347.0 | -4.0 | 320.0 | 48.6 | 3.0 | 279.4 | 26.0 |
| 6 | granulite | 5/3 | 47.9 | 68.8 | 47.2 | 345.0 | 5.4 | 39.9 | 69.9 | 48.0 | 340.3 | 7.3 |
| 7 | granulite | 5/5 | 52.4 | 78.9 | 28.2 | 334.8 | -7.5 | 18.8 | 81.5 | 31.4 | 323.2 | -5.5 |
| 8 | granulitic gneiss | 17/8 | 90.4 | 64.1 | 7.1 | 4.2 | -15.3 | 84.3 | 67.5 | 7.5 | 358.6 | -12.7 |
| 9 | granulitic gneiss | 13/12 | 48.9 | 62.6 | 7.8 | 351.5 | 11.0 | 42.3 | 69.4 | 8.9 | 342.0 | 7.3 |
| 10 | granulitic gneiss | 12/9 | 56.7 | 77.4 | 9.7 | 338.1 | -6.9 | 343.3 | 76.2 | 13.5 | 310.7 | 4.0 |
| 11* | granulitic gneiss | 18/17 | 52.1 | 17.9 | 3.0 | 22.7 | 30.6 | 40.9 | 25.6 | 5.7 | 10.8 | 37.0 |
| 12 | granulitic gneiss | 4/4 | 32.2 | 78.0 | 9.9 | 329.9 | -1.3 | 30.5 | 77.1 | 9.0 | 330.1 | 0.4 |
| 13 | granulitic gneiss | 10/3 | 31.9 | 61.6 | 20.8 | 342.4 | 19.7 | 26.0 | 61.3 | 21.7 | 338.5 | 22.1 |
| 14 | biotite-amphibole gneiss | 12/10 | 51.1 | 58.4 | 6.1 | 356.4 | 13.2 | 42.8 | 61.3 | 6.2 | 349.4 | 15.2 |
| 15 | granulitic gneiss | 12/8 | 68.0 | 75.4 | 11.8 | 343.8 | -9.0 | 30.6 | 81.3 | 11.2 | 326.7 | -6.2 |
| 16 | migmatitic gneiss | 11/11 | 51.3 | 67.6 | 5.1 | 348.0 | 5.4 | 52.2 | 67.4 | 5.6 | 348.6 | 5.1 |
| 22 | granulite | 9/3 | 125.6 | 63.7 | 27.3 | 5.8 | -39.4 | 129.6 | 63.9 | 25.8 | 4.7 | -42.1 |
| 23 | garnet granulite | 12/4 | 90.9 | 76.9 | 17.0 | 344.6 | -19.2 | 317.8 | 75.2 | 17.9 | 299.7 | 0.5 |
| 24 | orthogneiss | 10/6 | 53.0 | 73.1 | 12.7 | 342.9 | -0.6 | 39.1 | 76.5 | 9.5 | 334.2 | -0.3 |
| 26 | biotite gneiss | 8/7 | 64.8 | 76.0 | 3.4 | 342.5 | -8.2 | 316.3 | 79.3 | 9.7 | 304.2 | -5.5 |
| 27 | biotite gneiss | 4/4 | 341.1 | 84.0 | 27.2 | 314.5 | -9.8 | 301.8 | 78.4 | 28.2 | 299.3 | -8.4 |
| 29 | biotite gneiss | 7/6 | 10.0 | 70.6 | 3.5 | 324.0 | 13.4 | 331.7 | 46.8 | 8.2 | 288.0 | 33.6 |
| 34 | garnet-biotite-gneiss | 3/3 | 26.6 | 54.1 | 11.8 | 343.2 | 28.8 | 343.5 | 41.2 | 22.8 | 297.8 | 42.5 |
| 36 | mylonite | 3/3 | 277.8 | 71.0 | 9.1 | 283.2 | -13.0 | 289.6 | 78.3 | 2.4 | 296.7 | -12.3 |
| 38 | granulite | 3/3 | 354.6 | 57.5 | 25.5 | 313.5 | 30.6 | 328.8 | 56.2 | 24.1 | 291.1 | 25.1 |
| 40 | biotite-garnet-gneiss | 3/3 | 266.0 | 61.1 | 19.1 | 267.8 | -16.8 | 264.4 | 68.3 | 21.7 | 277.2 | -19.8 |
| 42 | orthogneiss | 6/6 | 34.1 | 47.5 | 6.4 | 353.2 | 30.4 | 21.4 | 51.1 | 18.4 | 340.1 | 33.3 |
| 45 | orthogneiss | 3/3 | 346.1 | 68.5 | 10.7 | 309.5 | 16.1 | 21.6 | 73.6 | 5.6 | 329.2 | 7.5 |
| 59 | biotite-gneiss | 3/3 | 320.9 | 58.8 | 45.0 | 287.4 | 19.2 | 303.6 | 73.0 | 43.0 | 292.6 | -2.1 |
| 66 | granulitic biotite-gneis | 3/3 | 100.8 | 58.7 | 27.9 | 12.9 | -21.3 | 39.5 | 73.4 | 35.7 | 337.4 | 3.4 |
| 70 | biotite-gneiss | 4/3 | 38.2 | 74.0 | 5.6 | 336.3 | 3.0 | 40.2 | 74.9 | 6.4 | 336.2 | 1.3 |
| 71 | granulitic gneiss | 3/3 | 164.8 | 85.2 | 3.2 | 321.3 | -30.3 | 202.2 | 81.4 | 3.4 | 310.6 | -36.5 |
| | Mean | 30 | 40.4 | 75.4 | 6.0 | 335.2 | -0.6 | 2.9 | 75.4 | 6.4 | 320.1 | 4.2 |
| | | | | | | $\alpha_{95}$(°) | K | | | | $\alpha_{95}$(°) | K |
| | | | | | | 10.0 | 7.9 | | | | 10.3 | 7.5 |

N/n-number of samples measured/number of samples used in the mean directions; Dec.-declination; Inc.-inclination; $\alpha_{95}$-Fisher's semi-angle confidence cone; K-Fisher's precision parameter; VGP-Virtual Geomagnetic Pole; Plong.-Paleolongitude; Plat.-Paleolatitude; AMS-Anisotropy of Magnetic Susceptibility; *-site 11 was not included in the mean.

magnetization as those of the Juiz de Fora and Piquete Complexes. Magnetic mineralogy and geochronological evidence suggests that these carbonatic rocks were remagnetized during the last phase of the Brasiliano orogeny (D'Agrella-Filho et al., 2000). It seems that remagnetization affected a large part of the stable area of the São Francisco Craton and adjacent fold belts at that time. Together, these facts suggest that no relative tilting movements occurred at least between these areas.

It is supposed that the Juiz de Fora Complex might have been affected by a late-collisional episode, between ~535 and 520 Ma, associated to sub-vertical dextral ductile shear zones, and retrogressive metamorphism described by Trouw et al. (2000). On the other hand, a K-Ar age of 499±18 Ma was determined in biotites from a pegmatite dyke that cuts granulite rocks from the Juiz de Fora Complex in the studied area, which was interpreted as the minimum age of the granulite rocks (Campanha, 1981). Therefore the age of the paleomagnetic pole for the Juiz de Fora Complex is probably between ~535–500 Ma.

For comparison, the RB and RBc poles are plotted in Figure 10 along with selected poles for the time interval between 560 and 500 Ma (Table 2), and their respective 95% confidence circles. The poles are rotated to the Gondwana fit of Lawver and Scotese (1987). Figure 10 shows an APW path for Gondwana as suggested by Meert et al. (2001). This curve provides the best fit for some new South American paleomagnetic poles (BGC, BGB, SF, SA1 and SA2, Table 2). The RB and RBc poles are statistically different at 95% confidence according to the F-test from MacFadden and Lowes (1981). The RBc pole

is located at an older (510 Ma) part of the APW path than RB (~500 Ma). These ages, however, are compatible with the range (535–500 Ma) suggested by radiometric ages from the Juiz de Fora Complex.

The PQ pole (Table 1, Fig. 10) was obtained from metamorphic rocks of the Piquete Complex (Ribeira belt, southwest of the studied area), and its K–Ar ages fall into the range 468–531 Ma (D'Agrella-Filho et al., 1986). Even though RB and PQ poles are statistically distinguishable at 95% confidence level, as indicated by the F-test (MacFadden and Lowes, 1981), they could have similar ages. Correction for AMS effect produced a shift of 16° in the original paleomagnetic pole from the Juiz de Fora Complex. A similar correction, if applied to the Piquete Complex paleomagnetic pole could produce a similar result.

Even though the geological evolution for the Juiz de Fora Complex based on geochronological data suggests an age in the range of ~535–500 Ma for the magnetization found in the studied rocks, the paleomagnetic results permit farther constraint on the age of this magnetization to 520–500 Ma by correlations with high quality paleomagnetic poles in the 560–500 Ma Gondwana APW path (Fig. 10).

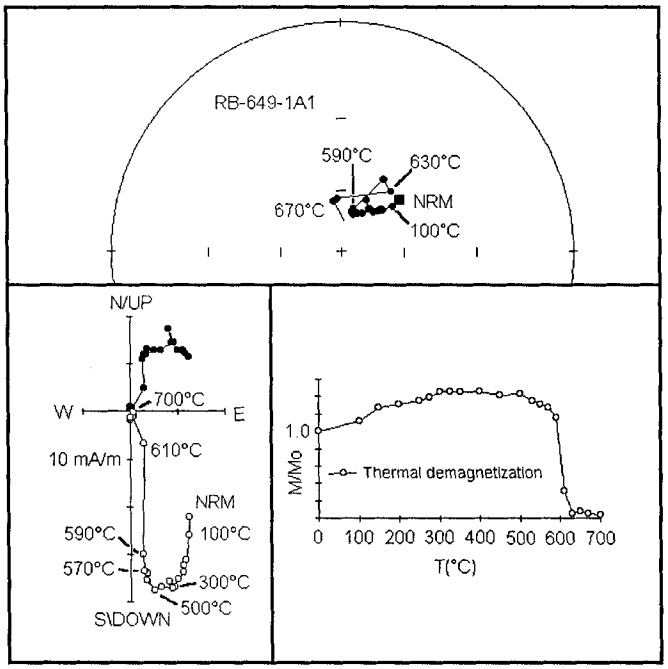



Fig. 6. Example of magnetic behavior for a pilot sample of the Juiz de Fora Complex. Equal-angle stereoplot, orthogonal vector diagram and normalized intensity decay plot. In the orthogonal vector diagram, open circles represent vectors projected onto the vertical plane and closed circles represent vectors projected onto the horizontal plane. In the stereoplot, open (closed) circles represent projections onto the upper (lower) hemisphere.

Table 2. Selected paleomagnetic poles from Gondwana (560–500 Ma).

| Pole | Plat. (°N) | Plong. (°E) | $\alpha_{95}$ (°) | Age (Ma.) | Reference |
|---|---|---|---|---|---|
| **African Poles** | | | | | |
| SD - Sinyai Dolerite | -28.4 | 319.1 | 5.0 | 547 | 1 |
| MB - Mirbat SS (Oman) | -31.9 | 333.9 | 7.2 | 550 | 2 |
| NR - Ntonya Ring Structure | 27.5 | 355.2 | 1.9 | 522 | 3 |
| **Antarctic Poles** | | | | | |
| SR - Sor Rondane | 10.6 | 8.3 | 4.5 | 515 | 4 |
| LK - Mt. Loke/Killer Ridge | 34.0 | 1.6 | 8.0 | 499 | 5 |
| **Australian Poles** | | | | | |
| CA1 - Mean pole # | -9.4 | 334.1 | 8.4 | 535 | 6 |
| CA2 - Mean pole # | 25.0 | 349.7 | 5.7 | 520 | 7 |
| CA3 - Mean pole # | 14.2 | 358.6 | 4.0 | 510 | 8 |
| **Indian Poles** | | | | | |
| BRM - Bhander-Rewa Mean | -23.0 | 333.0 | 11.0 | 550 | 9 |
| **Madagascan Poles** | | | | | |
| CR - Carion Granite | 12.7 | 359.7 | 11.0 | 508 | 10 |
| **South American Poles** | | | | | |
| SA2 - Sierra de las Animas | -35.3 | 307.0 | 8.0 | 560 | 11 |
| SA1 - Sierra de las Animas | 33.2 | 359.8 | 9.0 | 520 | 11 |
| BGB - Bambuí Gr. - B | 25.7 | 358.5 | 2.5 | 515 | 12 |
| BGC - Bambuí Gr. - C | 31.6 | 339.1 | 3.8 | 515 | 12 |
| SF - Salitre Fm. | 29.5 | 341.8 | 4.9 | 515 | 12 |
| PQ - Piquete Fm. | 23.0 | 22.0 | 10.2 | 510 | 13 |
| RB - Juiz de Fora Complex | 17.6 | 12.9 | 10.0 | 510 | 14 |
| RBc - Juiz de Fora Complex | 12.1 | 357.5 | 10.3 | 510 | 14 |
| **Sri Lanka Poles** | | | | | |
| TG - Tonigala Granite | 39.0 | 23.2 | 6.2 | 500 | 15 |

Plong.- paleolongitude; Plat. Paleolatitude; $\alpha_{95}$ Fisher's semi-angle confidence cone; 1- Meert and Van der Voo (1996); 2- Kempf et al. (2000); 3- Briden et al. (1993); 4- Zijderveld (1968); 5- Grunow and Encarnacion (2000); 6- mean calculated from the following poles: Lower Arumbela SS (Kirshvink, 1978), Brachina Fm. (McWilliams and McElhinny, 1980), Upper Arumbela SS (Kirshvink, 1978), Todd River Dolomite (Kirshvink, 1978); 7- mean calculated from the following poles: Bunyeroo Fm. (McWilliams and McElhinny, 1980), Hawker Gr. A (Klootwijk, 1980), Hawker Gr. B (Klootwijk, 1980); 8- mean calculated from the following poles: Aroona-Wirealpa-A (Klootwijk, 1980), Aroona-Wirealpa-B (Klootwijk, 1980), Tempe Fm. (Klootwijk, 1980), Hudson Fm. (Luck, 1972), Lake Frome-A (Klootwijk, 1980), Lake Frome-B (Klootwijk, 1980), Giles Creek Dol. – Lower (Klootwijk, 1980), Giles Creek Dol. – Upper (Klootwijk, 1980), Illara SS (Klootwijk, 1980), Deception Fm. (CA3) (Klootwijk, 1980); 9- McElhinny et al. (1978); 10-Meert et al. (2001); 1-Sánchez-Bettucci and Rapalini (2002); 12-D'Agrella-Filho et al. (2000); 13-D'Agrella-Filho et al. (1986); 14-this paper; 15-Yoshida et al. (1992) Poles were rotated to the Gondwana Fit of Lawver and Scotese (1987) -rotation parameters in Meert (2001).

## Final Remarks

Paleomagnetic analysis of high-grade metamorphic rocks (of varied lithotypes) from the Juiz de Fora Complex yielded a mean northeastern, steep downward inclination magnetization, carried mainly by (titano)magnetite and titanohematite, probably formed during the granulite facies metamorphism which affected these rocks at the final stages of the Brasiliano orogeny. This characteristic remanent magnetization (ChRM) was probably acquired during the uplift and cooling phase of this orogenic cycle

in the studied area, which probably occurred between 535–500 Ma. AMS measurements were used to correct the ChRM directions and consequently the paleomagnetic pole (RB) obtained for the analyzed rocks. The correction for the AMS effect on the mean ChRM produced a shift of ~9°. The paleomagnetic pole (RBc) calculated from the corrected directions is around 16° far from the one calculated from the uncorrected directions. These poles are statistically different at 95% confidence circle and the shift of 16° put the paleomagnetic pole closer to other high-quality poles that define an APW path for Gondwana in the time of 510 Ma.

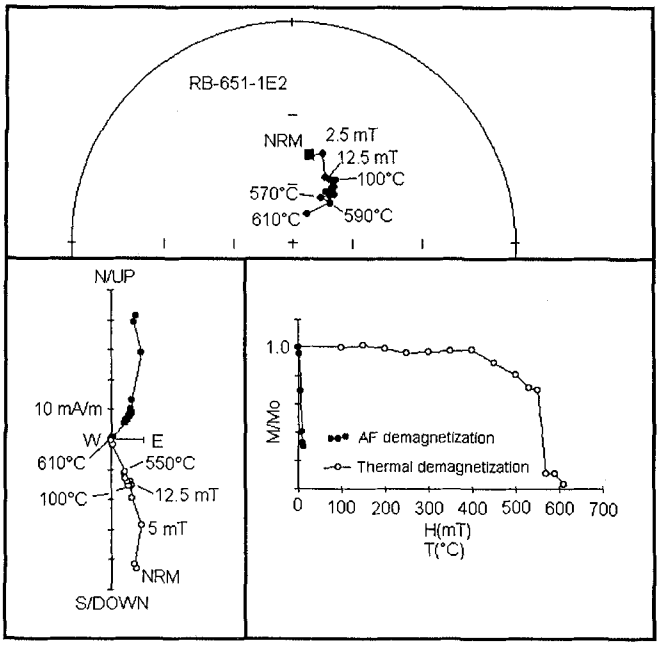



Fig. 7. Example of magnetic behavior for a pilot sample of the Juiz de Fora Complex as in figure 6.

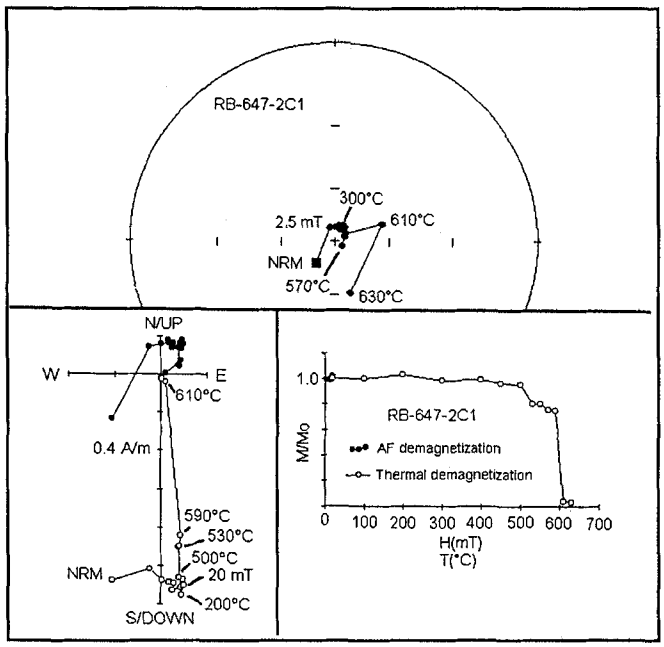



Fig. 8. Example of magnetic behavior for a pilot sample of the Juiz de Fora Complex as in figure 6.

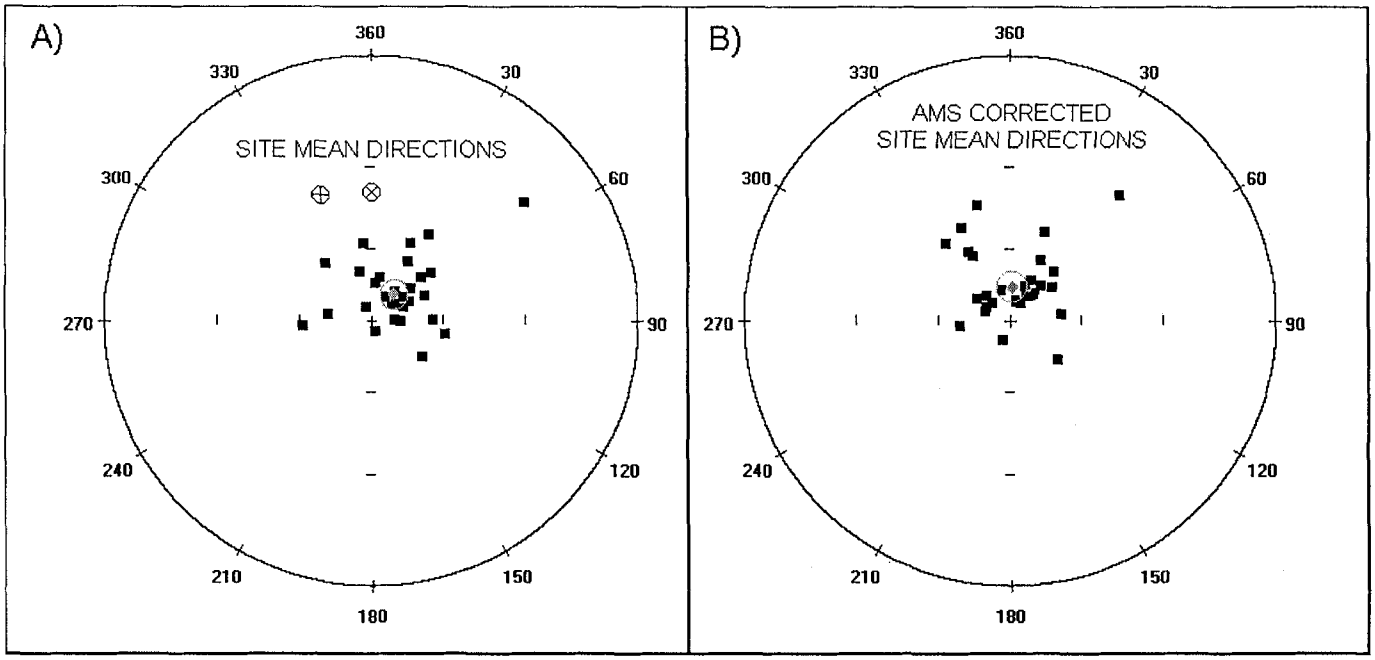



Fig. 9. (A) Site mean ChRM directions. ⊕ and ⊗ represent the actual geomagnetic field and the dipolar field, respectively. (B) Site mean ChRM directions corrected for the AMS effect. The mean directions and $\alpha_{95}$ circles calculated for the Juiz de Fora Complex for the uncorrected (A) and corrected (B) AMS effect are also represented (light gray). Symbols are the same as in figure 6.

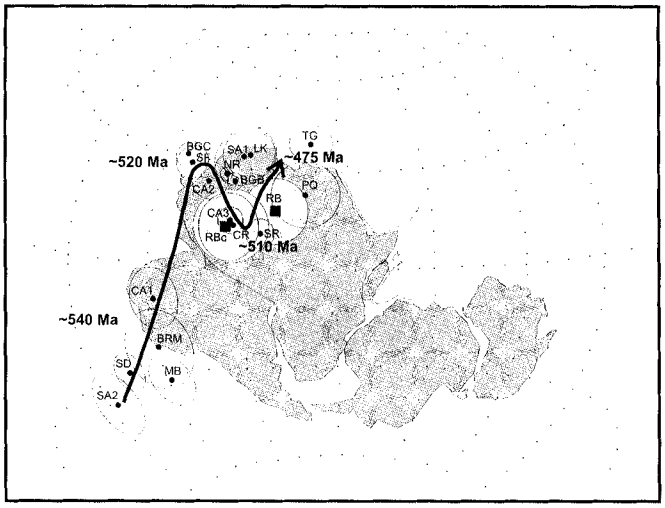



Fig. 10. Gondwana paleomagnetic poles from table 2 in the range 560-500 Ma rotated to the Gondwana configuration of Lawver and Scotese (1987) with the African poles fixed (rotation parameters in Meert et al., 2001). Squares represent the paleomagnetic poles calculated for the Juiz de Fora Complex (for uncorrected (RB) and corrected AMS effect (RBc)). The APW path for Gondwana is similar to that suggested by Meert et al. (2001).

This paper shows that high-grade metamorphic rocks can also be suitable for paleomagnetic studies, at least for the rocks from the Juiz de Fora Complex. Also, the AMS measurements can be used as a first approximate correction of the ChRM directions affected by magnetic anisotropies.

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## References

Bascou, J., Raposo, M.I.B., Vauchez, A. and Egydio-Silva, M. (2002) Titanohematite lattice preferred orientation and magnetic anisotropy in high-temperature mylonites. Earth Planet. Sci. Lett., v. 198, pp. 77-92.

Briden, J.C., McClelland, E. and Rex, D.C. (1993) Proving the age of a paleomagnetic pole: the case of the Ntonya Ring Structure, Malawi. J. Geophys. Res., v. 98, pp. 1743-1749.

Brito Neves, B.B. de, Campos Neto, M.C. and Fuck, R.A. (1999) From Rodinia to Western Gondwana: an approach to the Brasiliano-Pan African cycle and orogenic collage. Episodes, v. 22, pp. 155-166.

Campanha, G.A.C. (1981) O lineamento de Além-Paraíba na área de Três Rios (RJ). Rev. Brasil. Geoc., v. 11, pp. 159-171.

D'Agrella-Filho, M.S., Babinski, M., Trindade, R.I.F., Van Schmus, W.R. and M. Ernesto, M. (2000) Simultaneous remagnetization and U-Pb isotope resetting in Neoproterozoic carbonates of the São Francisco craton, Brazil. Precambrian Res., v. 99, pp. 179-196.

D'Agrella-Filho, M.S., Trindade, R.I.F., Siqueira, R., Ponte-Neto, C.F. and Pacca, I.G. (1998) Paleomagnetic constraints on the Rodinia Supercontinent: implications for its Neoproterozoic break-up and the formation of Gondwana. International Geology Rev., v. 40, pp. 171-188.

D'Agrella-Filho, M.S., Pacca, I.G. and Sato, K. (1986) Paleomagnetism of metamorphic rocks from the Piquete Region - Ribeira Valley, Southeastern Brazil. Rev. Brasil. Geof., v. 4, pp. 79-84.

Dalziel, I.W.D. (1991) Pacific margin of Laurentia and East Antarctica/ Australia as a conjugate rift pair: evidence and implications for an Eocambrian supercontinent. Geology, v. 19, pp. 586-601.

Dalziel, I.W.D. (1997) Neoproterozoic-Paleozoic geography and tectonics: review, hypothesis. environmental speculation. Bull. Geol. Soc. Amer., v. 109, pp. 16-42.

Duarte, B.P., Heilbron, M. and Campos Neto, M.C. (2000) Granulite/charnockite from the Juiz de Fora domain, central segment of the Brasiliano Ribeira Belt. Rev. Bras. de Geoc., v. 30, pp. 358-362.

Egydio-Silva, M., Vauchez, A., Bascou, J. and Hippert, J. (2002) High-temperature deformation in the Neoproterozoic transpessional Ribeira belt, southeast Brazil. Tectonophys., v. 352, pp. 203-224.

Evans, D.A.D., Li, Z.X., Kirschvink, J.L. and Wingate, M.T.D. (2000) A high-quality mid-Neoproterozoic paleomagnetic pole from South China, with implications for ice ages and the breakup configurations of Rodinia. Precambrian Res., v. 100, pp. 313-334.

Fisher, R.A. (1953) Dispersion on a sphere. Proc. R. Soc., Ser. v. A 217, pp. 295-305.

Grunow, A.M. and Encarnacion, J. (2000) Cambro, Ordovician paleomagnetic and geochronologic data from southern Victoria Land, Antarctica: revision of the Gondwana apparent polar wander path. Geophys. J. Int., v. 141, pp. 391-400.

Hoffman, P.F. (1991) Did the breakout of Laurentia turn Gondwanaland inside-out? Science, v. 252, pp. 1409-1412.

Karsltrom, K.E., Harlan, S.S., Williams, M.L., Mclelland, J., Geissman, J.W. and Ahäll, K.-I. (1999) Refining Rodinia: Geologic evidence for the Australia- Western U.S. connection in the Proterozoic. GSA Today, v. 9, pp. 1-7.

Kempf, O., Kellerhalls, P., Lowrie, W. and Matter, A. (2000) Paleomagnetic directions in Precambrian/Cambrian glaciomarine sediments of the Mirbat sandstone, Oman. Earth Planet. Sci. Lett., v. 175, pp. 181-190.

Kirshvink, J.L. (1978) The Precambrian-Cambrian boundary problem: Paleomagnetic directions from the Amadeus basin, Central Australia. Earth Planet. Sci. Lett., v. 40, pp. 91-100.

Kirshvink, J.L. (1980) The least-saquares line and plane and the analysis of paleomagnetic data. Geophys. J. R. Astr. Soc., v. 62, pp. 699-718.

Klootwijk, C.T. (1980) Early Paleozoic paleomagnetism in Australia. Tectonophys., v. 64, pp. 249-332.

Lawver, L. and Scotese, C.R. (1987) A revised reconstruction of Gondwanaland. In: McKenzie, G.D. (Ed.), Gondwana Six: structure, tectonics and geophysics. Amer. Geophys. Un. Mon., v. 40, pp. 17-23.

Luck, G.R. (1972) Paleomagnetic results from Paleozoic sediments of N. Australia. Geophys. J. Roy. Astr. Soc., v. 28, pp. 475-487.

Machado, N., Valladares, C., Heilbron, M. and Valeriano, C. (1996) U–Pb geochronology of the central Ribeira belt (Brazil) and implications for the evolution of the Brazilian Orogeny. Precambrian Res., v. 79, pp. 347-361.

McElhinny, M.W., Cowley, J.A. and Edwards, D. (1978) Paleomagnetism of some rocks from peninsular India and Kashmir. Tectonophys., v. 50, pp. 649-687.

McFadden, P.L. and Lowes, F.J. (1981) The discrimination of mean directions drawn from Fisher distributions. Geophys. J. R. Astr. Soc., v. 67, pp. 19-33.

McWilliams, M.O. and McElhinny, M.W. (1980) Late Precambrian paleomagnetism in Australia in the Adelaide geosyncline. J. Geol., v. 88, pp. 1-26.

Meert, J.G. (2001) Growing Gondwana and rethinking Rodinia: a paleomagnetic perspective. Gondwana Res., v. 4, pp. 278-288.

Meert, J.G. and Van der Voo, R. (1996) Paleomagnetic and $^{40}Ar/^{39}Ar$ study of the Sinyai dolerite, Kenya: implications for Gondwana assembly. J. Geol., v. 104, pp. 131-142.

Meert, J.G. and Van der Voo, R. (1997) The assembly of Gondwana 800-550 Ma. J. Geodyn., v. 23, pp. 223-235.

Meert, J.G., Nédélec, Hall, A. C. Wingate, M.T.D. and Rakotondrazafy, M. (2001) Paleomagnetism, geochronology and tectonic implications of the Cambrian-age Carion Granite, Central Madagascar. Tectonophys., v. 340, pp. 1-21.

Meert, J.G., Van der Voo, R. and Ayub, S. (1995) Paleomagnetic investigation of the Neoproterozoic Gagwe lavas and Mbozi complex, Tanzania and the assembly of Gondwana. Precambrian Res., v. 74, pp. 225-244.

Raposo, M.I.B. and Egydio-Silva, M. (2001) Magnetic fabric studies of high-grade metamorphic rocks from the Juiz de Fora Complex, Ribeira belt, SE Brazil. Intern. Geol. Rev., v. 43, pp. 441-456.

Raposo, M.I.B., D'Agrella-Filho, M.S. and Siqueira, R. (2003) The effect of magnetic anisotropy on paleomagnetic directions in high-grade metamorphic rocks from the Juiz de Fora Complex, SE Brazil. Earth Planet. Sci. Lett., v. 209, pp. 131-147.

Rogers, J.J.W., Unrug, R. and Sultan, M. (1995) Tectonic assembly of *Gondwana*. J. Geodyn., v. 19, pp. 1-34.

Sánchez-Bettucci, L. and Rapalini, A.E. (2002) Paleomagnetism of the Sierra de Las Animas Complex, southern Uruguay: its implications in the assembly of western Gondwana. Precambrian Res., v. 118, pp. 243-265.

Schmitt, R.S., Trouw, R.A.J. and Van Schmus, W.R. (1999) The characterization of a Cambrian (520 Ma) tectometamorphic event in the coastal domain of the Ribeira Belt (SE Brazil) – using U/Pb in syntectonic veins. In: Actas II South American Symp. on Isotope Geology, Cordoba, Argentina, Extended Absts., pp. 363-366.

Tauxe, L., Mullender, T.A.T. and Pick, T. (1996) Potbellies, wasp-waists, and superamagnetism in magnetic hysteresis. J. Geophys. Res., v. 101, pp. 571-583.

Torsvik, T.H., Carter, L.M., Ashwal, L.D., Bhushan, S.K., Pandit, M.K. and Jamtveit, B. (2001) Rodinia refined or obscured: palaeomagnetism of the Malani igneous suite (NW Indian). Precambrian Res., v. 108, pp. 319-333.

Trompette, R. (1994) Geology of Western Gondwana (2000–500 Ma). Pan-African-Brasiliano aggregation of South America and Africa. Rotterdan, Balkema, 1994, 350 p.

Trouw, R., Heilbron, M., Ribeiro, A., Paciullo, F., Valeriano, C.M., Almeida, J.C.H., Tupinambá, M. and Andreis, R.R. (2000) The central segment of the Ribeira Belt. In: Cordani, U.G., Milani, E.J., Thomaz-Filho, A. and Campos, D.A. (Eds.), Tectonic evolution of South America, Rio de Janeiro, Brazil pp. 287-310

Tupinambá, M., Teixeira, W., Heilbron, M. and Basei, M.A.S. (1998) The Pan-African/Brasiliano arc-related magmatism at the Costeiro domain of the Ribeira belt, southeastern Brazil: new geochronological and lithogeochemical data. In: 14th International conference on basement tectonics. Ouro Preto, MG, Brazil, Extended Abst., pp. 12-14.

Weil, A.B., Van der Voo, R., Niocaill, C.M. and Meert, J.G. (1998) The Proterozoic supercontinent Rodinia: paleomagnetically derived reconstructions for 1100 to 800 Ma. Earth Planet. Sci. Lett., v. 154, pp. 13-24.

Wingate, M.T.D. and Giddings, J.W. (2000) Age and palaeomagnetism of the Mundine Well dyke swarm, Western Australia: implications for an Australia-Laurentia connection at 755 Ma. Precambrian Res., v. 100, pp. 335-357.

Yoshida, M., Funaki, M. and Vitanage, P.W. (1992) Proterozoic to Mesozoic East Gondwana: the juxtaposition of India, Sri Lanka and Antarctica. Tectonics, v. 11, pp. 381-391.

Zijderveld, J.D.A. (1967) A. C. Demagnetization of rocks: analysis of results. In: Collinson, D.W., Creer, K.M. and Runcorn, S.K. (Eds.), Methods in paleomagnetism, Elsevier, Amsterdam, pp. 254-286.

Zijderveld, J.D.A (1968) Natural remanent magnetization of some intrusive rocks from the SØr Rondane Mountains, Queen Maud Land, Antarctica. J. Geophys. Res., v. 73, pp. 3773-3785.